

Amit Kumar

Senior Data Scientist • Agentic AI Engineer • GenAI Engineer

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Summary

Senior Data Scientist, GenAI Engineer, and Agentic AI Developer with 4.6+ years of experience designing and deploying production AI systems across automobile, education, transport, and retail domains. Architected multi-agent AI platforms using LangGraph and LangChain, orchestrating 8–11 specialized agents for autonomous workflow execution and intelligent decision-making. Delivered enterprise-scale RAG systems, LLM-powered applications, computer vision solutions, and on-premise AI deployments. Proven track record of reducing manual processes from days to minutes, achieving over 95% model accuracy, and building scalable AI platforms using FastAPI, AWS, Docker, OpenAI, FAISS, and modern LLM frameworks.

Experience

Senior Data Scientist | Agentic AI
GlobalNodes, Gurugram, India

Jul 2025 – Present

- Architected an AI-native QA automation platform using LangGraph and LangChain, orchestrating 8–11 specialized subagents for NLP-to-test conversion, automated test data generation, framework-aware code generation, and CI/CD integration.
- Delivered 10× faster testing workflows, >99% potential test coverage, and reduced time-to-first-test to under 4 minutes from a plain-language instruction.
- Enabled end-to-end test lifecycle automation: natural language input → test case generation → execution → structured report output.
- Built FastAPI backend with JWT authentication, Role-Based Access Control (Super Admin / Admin / User), and MySQL ORM for workflow persistence, execution history, and audit logs.
- Implemented RAG pipelines using FAISS with OpenAI Embeddings and SentenceTransformers for context-aware, codebase-grounded test generation.
- Designed memory persistence, tool-calling workflows, structured outputs, and execution guardrails for reliable production behavior across 15+ intelligent platform capabilities.

Associate AI/ML → Senior Associate AI/ML
Unthinkable Solutions, Gurugram, India

Nov 2021 – Jul 2025

- Delivered end-to-end AI/ML solutions across automobile, education, transport, and retail domains; evolved from computer vision to GenAI/LLM-powered applications.
- Built a fracture analysis system for an automotive manufacturing client, reducing manual inspection time from **2 days to under 10 minutes** at >95% accuracy – deployed on-premise at client site.
- Developed deep learning models (TensorFlow, PyTorch) for object detection, classification, and segmentation across 5+ client projects in computer vision.
- Built and deployed LangChain-based AI workflows and retrieval-driven applications for enterprise use cases.
- Delivered scalable ML pipelines on AWS using SageMaker, EC2, Docker, and REST services in production environments.
- Collaborated with data engineering teams to optimize ETL pipelines and streamline production deployments.

Key Projects

AI-Native QA Automation Platform

GlobalNodes, 2025–Present

- Multi-agent system with 8–11 subagents handling NLP-to-test conversion, test data generation, CI/CD integration, and automated reporting – reducing time-to-first-test to under 4 minutes.
- Implemented JWT authentication and RBAC supporting three access tiers; MySQL ORM for workflow persistence, execution history, and session management.
- Designed asynchronous FastAPI backend with modular architecture, tool-calling pipelines, and structured output schemas for reliable agentic behavior.

Fracture Analysis Assistant

Automotive Manufacturing Client, On-Premise Deployment

- Built an AI-powered fracture analysis system covering 15+ polymer types and complex multi-part assemblies, replacing a 2-day manual process with automated analysis in under 10 minutes.
- Trained deep learning classification models achieving >95% accuracy (~94% precision, ~93% recall) across Brittle, Ductile, and Negative fracture classes.
- Applied targeted data augmentation and model optimization to ensure robustness across diverse manufacturing scenarios; deployed on-premise on client infrastructure.

Car & Accessory Design – 3D GenAI

Internal Tool, Production Use

- Fine-tuned LLMs to generate Python-based CadQuery and Blender scripts from natural language, enabling engineers to go from design intent to a working 3D CAD file in a single prompt.
- Reduced prototyping iteration time by eliminating manual scripting – first-draft 3D models now delivered as usable CAD/Blender files directly from text instructions.
- Integrated open-source text-to-3D models, customized for automotive domain-specific components; actively used in internal design workflows.

Skills

Agentic AI & Multi-Agent Systems: Agentic AI, Multi-Agent Architectures, LangGraph, LangChain, AI Agents, Workflow Orchestration, Tool Calling, Function Calling, Structured Outputs, Memory Systems, Human-in-the-Loop Workflows, Agent Planning, Autonomous Task Execution

Generative AI & LLMs: Large Language Models (LLMs), Prompt Engineering, Prompt Optimization, Retrieval-Augmented Generation (RAG), AI provider APIs, Embeddings, Semantic Search, Context Engineering, LLM Evaluation

LLM Optimization & Inference: Prompt Tuning, Inference Optimization, Latency Optimization, Token Optimization, Model Evaluation, Cost Optimization

Retrieval & Vector Databases: FAISS, Vector Search, Embedding Models, Similarity Search, Hybrid Search, Knowledge Retrieval

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Computer Vision, Deep Learning, Classification, Object Detection, Segmentation, Feature Engineering, Model Training, Model Evaluation

Backend & AI Platforms: Python, FastAPI, REST APIs, Async Processing, Microservices, MySQL, ORM, JWT Authentication, Role-Based Access Control (RBAC), API Design

MLOps & Production AI: Model Deployment, Model Monitoring, Data Drift Detection, Performance Monitoring, CI/CD Pipelines, Experiment Tracking, Continuous Learning Pipelines, Feedback Loops, Production ML Lifecycle Management

Cloud, Infrastructure & Distributed Systems: AWS (SageMaker, EC2, S3), Docker, Docker Compose, Linux, Git, Distributed Systems, Scalable AI Infrastructure, High-Throughput Inference Systems, On-Premise AI Deployments

Data & Analytics: SQL, Pandas, NumPy, Data Processing, ETL Pipelines, Data Validation, Data Quality Monitoring

Education

Bachelor of Technology in Computer Science
Meerut Institute of Engineering and Technology

2018 – 2022
CGPA: 7.98